Urban Sound Classification

Road Traffic and Acoustic Events

User Guide

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Pipeline Overview

The project is organized as a sequential pipeline of eight Jupyter notebooks. Each notebook produces files consumed by the next. The full pipeline runs in order from 01 to 08; individual stages can be skipped if their outputs already exist on disk.

NB	Name	Description	Runtime (CPU)
01	Data Exploration	Inspect source databases, check labels, durations, sample rates	<1 min
02	Segmentation Pipeline	Convert all raw audio to 5s mono WAV clips at 22 050 Hz	5-15 min
03	Feature Extraction	Compute log-mel and PCEN spectrograms, save as .npy	10-20 min
04	Dataset and DataLoader	File-level train/val/test split, compute normalisation stats	1-5 min
05	Model Training	Train a 2D CNN with SpecAugment and early stopping	30-60 min
06	24h Acoustic Analysis	Sliding-window inference on field recordings, 14 figures	5-15 min
07	Top Noises Graph	Class distribution pie charts and 24h event timeline from NB 06 outputs	<5 min
08	t-SNE Visualisation	256-d CNN embeddings reduced to 2D, 5 figures across 10 classes	5-15 min

Prerequisites

Software

Python 3.10 or later is required. Dependencies can be installed in either of the following ways.

Option A: Install packages manually

```
pip install torch torchvision torchaudio pytorch-lightning
pip install librosa soundfile numpy pandas matplotlib seaborn scikit-learn tqdm
    ipywidgets
```

Option B: Create a virtual environment and install from requirements.txt

```
python -m venv .venv

# Windows
.venv\Scripts\activate

# macOS / Linux
source .venv/bin/activate

pip install -r requirements.txt
```

Tested with: Python 3.10, PyTorch 2.1, pytorch-lightning 2.1, librosa 0.10, numpy 1.26.

Required External Datasets

Three datasets must be obtained and placed in the `data/` directory before running notebook 01.

Dataset	Classes	Source
IDMT-Traffic	Car, MC, Bus, Truck, BG	Fraunhofer IDMT website
Melaudis	Car, MC, Bus, Truck, Tram, BG	Contact the Melaudis project team
UrbanSound8K	Construction, Dog, Horn, Siren	`urbansounddataset.weebly.com`

Field Recordings (Notebooks 06 and 07 only)

Notebooks 06 and 07 expect continuous audio files covering a 24-hour period, split into three folders:

Folder	Period	Hours
`audios_16h_00h/`	Evening	16:00 to 00:00
`audios_00h_08h/`	Night	00:00 to 08:00
`audios_08h_16h/`	Day	08:00 to 16:00

Step-by-Step Usage

Option A: Full pipeline from scratch

Use this option when starting from raw datasets with no pre-computed files.

```
jupyter notebook 01_data_exploration.ipynb        # no files written
jupyter notebook 02_segmentation_pipeline.ipynb   # produces data/segments/
jupyter notebook 03_feature_extraction.ipynb      # produces data/features/
jupyter notebook 04_dataset_dataloader.ipynb      # produces norm_stats + split
jupyter notebook 05_model_training.ipynb          # produces checkpoints/
jupyter notebook 06_24h_acoustic_analysis.ipynb   # produces figures + summary CSVs
jupyter notebook 07_topnoisesgraph.ipynb          # produces top-noise figures
    (requires NB 06 outputs)
jupyter notebook 08_tsne.ipynb                    # produces embeddings + t-SNE
    figures (requires NB 03-05 outputs)
```

Option B: Resume from pre-computed features

Use this option if `data/features/` and `data/features_manifest.csv` already exist.

```
jupyter notebook 04_dataset_dataloader.ipynb
jupyter notebook 05_model_training.ipynb
jupyter notebook 06_24h_acoustic_analysis.ipynb
jupyter notebook 07_topnoisesgraph.ipynb
jupyter notebook 08_tsne.ipynb
```

Option C: Use an existing checkpoint

Use this option if `checkpoints/logmel_best-v1.ckpt` already exists and you only need the analysis and further graphs.

```
jupyter notebook 06_24h_acoustic_analysis.ipynb
jupyter notebook 07_topnoisesgraph.ipynb          # requires NB 06 outputs
jupyter notebook 08_tsne.ipynb                    # requires checkpoints/ +
    data/features/
```

Checkpoint reuse: the checkpoint file contains the full model weights. Notebooks 01 to 04 do not need to be re-run to perform inference. Notebook 08 loads the checkpoint directly to extract embeddings.

Key Configuration Parameters

All parameters are defined at the top of each notebook under **Section 0: Imports & configuration**. The table below lists the values most likely to require adjustment.

Parameter	Default	Notebook	Description
`TARGET_SR`	22 050	02	Output sample rate (Hz)
`CAP`	1 000	02	Maximum clips per class
`SEED`	42	02, 04	Random seed (reproducibility)
`N_MELS`	64	03	Number of mel filterbank bins
`FEATURE_TYPE`	logmel	04, 05, 08	Feature to use (logmel or pcen)
`BATCH_SIZE`	64	04, 05	DataLoader batch size
`LR`	1e-3	05	Initial learning rate
`MAX_EPOCHS`	60	05	Training epoch ceiling
`PATIENCE`	10	05	Early stopping patience